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TotalPlant Solutions (TPS) System

Fieldbus
TPS Fieldbus
Reference

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TotalPlant

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Honeywell
Industrial Automation and Control
16404 North Black Canyon Highway
Phoenix, AZ 85053
1-800 343-0228

About This Document

This document describes the Fieldbus system requirements and supplies references for,

- Function block user applications
- Alarms
- Fieldbus Control
- HPM Fieldbus point display
- FF Status Byte Display
- Parameters cross reference

Release Information

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Contacts

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Organization		Phone Number	
United States and Canada	Honeywell Inc.	1-800-343-0228	Sales Service Tech. Support
	Industrial Automation and Control	1-800-525-7439	
		1-800-423-9883	
Asia Pacific	Honeywell Asia Pacific Inc. Hong Kong	(852) 28298298	
Europe	Honeywell PACE Brussels, Belgium	[32-2] 728-2111	
Latin America	Honeywell Inc. Sunrise, Florida U.S.A.	(954) 845-2600	

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

About This Document
Symbol Definitions

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Contents

1. System Requirements and Specifications

1.1 Platform Requirements

Hardware Platform

The minimum TPS hardware requirements for fieldbus are listed in the following table.

Hardware Component	Notes
Global User Station	Intel version with 128 MB RAM, FS
Primary Domain Controller	(NT) Server required by TPS Builder, if not already present
TotalPlant Network Server	TPN
History Module	
Network Interface Module	NIM
High-Performance Process Manager	HPM
Fieldbus I/O Processors	IOPs
Fieldbus Field Termination Assemblies	FTAs
Field Devices	FOUNDATION Fieldbus certified fieldbus device(s)
NI Configurator card	Installed in a PC other than the GUS which will be used for downloading



ATTENTION

Fieldbus may be configured on a workstation with NT Service Pack 4. The downloading of the fieldbus configuration requires a Global User Station with TPS Builder (Release 120). After fieldbus is configured, a Global User Station or a Universal Station can be used for operating Fieldbus points.

System Requirements and Specifications

Platform Requirements

Software Platform

Fieldbus utilizes the following required and optional software.

Software Component	Status	Notes
Honeywell R610.xx with NT 4.0, Service Pack 4 or later	Required	
TPS Builder Release 120	Required	
File Transfer	Required	Installed on the GUS
Global User Station Release 203.4 or later	Required	
NI Configurator	Required	Software used to download to device firmware
Microsoft Excel or Excel 97	Required	Used by TPS Builder to load data into spreadsheets
Premia Codewright Professional 5.1	Required	Supplied with the LCN/UCN library of TPS Builder
Seagate Crystal Reports Pro 5.097	Optional	Reports are created using Crystal Reports; required only if you intend to edit reports.
Adobe Acrobat 3.0 97	Required	Used to access some Honeywell documentation and supplied with TPS Builder
OLE Automation controller	Required	Any OLE Automation controller, such as Microsoft Visual Basic 5.0 or Microsoft Excel 7.0 97
Microsoft Access 7.0 or other ODBC compliant application	Required	An Access-type database structure is used, but the engine need not be present.
ODBC Data Source Administration 97	Optional	Used for compacting data. While not absolutely necessary, its use is encouraged to conserve space

1.2 Fieldbus IOP and FTA Specifications

Parameter	Specification
IOP Model Numbers	MU-PFBS01, MC-PFBS01
FTA Model Numbers	MU-TFBS01, MC-TFBS01 • Div 2 with power supply modules (MTL4995) • Compression Terminals MU-TFBS11, MC-TFBS11 • Div 2 without power supply modules • Compression Terminals MU-GFBS01, MC-GFBS01 • IS with power supply / isolator modules (MTL4053) • Compression Terminals
Physical Interface	H1 FOUNDATION Fieldbus
Devices Supported	Multi-vendor Qualified FOUNDATION Fieldbus Compatible Devices
Number of Devices per IOP/ FTA	4 Segments per IOP/FTA; up to 10 devices per segment: • Current limited to 350mA for Div 2 using MTL4995 • Current limited to 80mA for IS using MTL4053

System Requirements and Specifications

Fieldbus IOP and FTA Specifications

Parameter	Specification
Distances	IOP to FTA: • Internal cable within HPM cabinet • External cable 50 m maximum HPM Power Subsystem to Power Distribution Assembly: • Internal cable within HPM cabinet • External cable 50 m maximum • Power Distribution Assembly to FTA: • Internal cable within cabinet only FTA to Field Devices: • Per Clause 22 of IEC 1158-22 specification • IS distance limited by cable impedance (refer to MTL Application Brief AB003 for more details)
Power Requirements	FOUNDATION Fieldbus IOP: • 700mA typical @ 24V • 1.25A max. @ 24V Div 2 FTA: • 960mA typical @ 24V in and 200mA output current/segment • 2A max. @ 20V in and 350mA output current/segment IS FTA: • 540mA typical @ 24V in and 80mA output current/segment
Surge withstand capability	ANSI/IEEE C37.90.1-1978
ESD Protection	IEC 801.2
FTA Isolation	250 VAC between safe and hazardous area circuits
Environmental Conditions	Same as PM family
Electromagnetic Interference	15 V/m (External to the standard HPM cabinet with doors closed.)

System Requirements and Specifications
Fieldbus IOP and FTA Specifications

Parameter	Specification
Surge Protection	IEEE SWC 472-1974
CE Conformity (Europe)	89/336/EEC, Electromagnetic Compatibility (EMC) Directive
LED Indicators	IOP: • 2 LEDs – indicate power and module status (green) FTA: • 4 LEDs (one per power supply/isolator) - indicate power to modules (green)
Major Dimensions (FTA)	4.9" W x 12.1" L (B-size)
GI/IS Power Assembly & Cables Power Distribution Panel Power Cable (Local Cabinet Use) Power Cable	MU-GPRD02, MC-GPRD02 (Galvanic Isolation) MU-KGPR00 MU-KSPRxx (where xx is the length in meters as follows: 05, 10, 20, 25, 30, 35, 40, 45, or 50)
FTA I/O Cables FTA I/O Cable FTA I/O Cable (CE Compliant IOP)	MU-KFTAxx (where xx is the length in meters as follows: 05, 10, 20, 25, 30, 35, 40, 45, or 50) MU-KFTSxx (where xx is the length in meters as follows: 05, 10, 20, 25, 30, 35, 40, 45, or 50)

System Requirements and Specifications

Fieldbus IOP and FTA Specifications

2. Function Block User Application Reference

2.1 Block Parameters Summary

This table provides a description of in the information provided in Block Parameter Summary.

Column	Description
AI Idx	Index number into the Analog Input function block parameter list.
Parameter	FF Parameter name
OT	Object type: S = Simple Variable R = Record A = Array of simple variables
DT	Data Type: - US8, US16, US32 = unsigned variables of 8, 16 or 32 bits. - F = Floating point - BS = Bit string - VS = Visible String - OS = Octet String - DS = Data Structures Number following is the FF Data Structure Type, which may be: - Value and status – float – value and status of a floating point parameter - Scaling – Static data used to scale floating point values for display purposes.
USE	C = Contained; I = function block Input; O = function block Output.
Stores	Indicates type of memory where parameter is stored: - S – Static Writing to the parameter increments the static revision counter ST_REV. - N – Non-volatile Parameter must be retained during a power cycle. It is not under the static update code.

Column	Description
	D – Dynamic The value is calculated by the block or read from another block.
Size	The number of octets.
VIEW_1,2,3,4	Visibility in the VIEW.

2.2 Resource Block Parameters

The following table lists the FF Resource Block Parameters.

RB Idx	Parameter Name	OT	Data Type	Use	Model	VIEW_x			
						1	2	3	4
1	ST_REV	S	US16	C	Stat Revision	Yes	Yes	Yes	Yes
2	TAG_DESC	S	OS	C	Tag Desc	No	No	No	No
3	STRATEGY	S	US16	C	Strategy	No	No	No	Yes
4	ALERT_KEY	S	US8	C	Alert Key	No	No	No	Yes
5	MODE_BLK	R	DS69	C	Mode	Yes	No	Yes	No
6	BLOCK_ERR	S	BS	C	Block Error	Yes	No	Yes	No
7	RS_STATE	S	US8	C	Resource State	Yes	No	Yes	No
8	TEST_RW	R	DS85	C	Test	No	No	No	No
9	DD_RESOURCE	S	VS	C	DD Resource	No	No	No	No
10	MANUFAC_ID	S	US32	C	Contained	No	No	No	Yes
11	DEV_TYPE	S	US16	C	Contained	No	No	No	Yes
12	DEV_REV	S	US16	C	Contained	No	No	No	Yes
13	DD_REV	S	US8	C	Contained	No	No	No	Yes
14	GRANT_DENY	R	DS70	C	Access Perm	No	Yes	No	No

Function Block User Application Reference
Resource Block Parameters

RB Idx	Parameter Name	OT	Data Type	Use	Model	VIEW_x			
						1	2	3	4
15	HARD_TYPES	S	BS	C	Contained	No	No	No	Yes
16	RESTART	S	US8	C	Contained	No	No	No	No
17	FEATURES	S	BS	C	Contained	No	No	No	Yes
18	FEATURE_SEL	S	BS	C	Contained	No	Yes	No	No
19	CYCLE_TYPE	S	BS	C	Contained	No	No	No	Yes
20	CYCLE_SEL	S	BS	C	Contained	No	Yes	No	No
21	MIN_CYCLE_T	S	US32	C	Contained	No	No	No	Yes
22	MEMORY_SIZE	S	US16	C	Contained	No	No	No	Yes
23	NV_CYCLE_T	S	US32	C	Contained	No	Yes	No	No
24	FREE_SPACE	S	F	C	Contained	No	Yes	No	No
25	FREE_TIME	S	F	C	Contained	Yes	No	Yes	No
26	SHED_RCAS	S	US32	C	Contained	No	Yes	No	No
27	SHED_ROUT	S	US32	C	Contained	No	Yes	No	No
28	FAULT_STATE	S	US8	C	Contained	Yes	No	Yes	No
29	SET_FSTATE	S	US8	C	Contained	No	No	No	No
30	CLR_FSTATE	S	US8	C	Contained	No	No	No	No
31	MAX_NOTIFY	S	US8	C	Contained	No	No	No	Yes
32	LIM_NOTIFY	S	US8	C	Contained	No	Yes	No	No
33	CONFIRM_TIME	S	US32	C	Contained	No	Yes	No	No
34	WRITE_LOCK	S	US8	C	Contained	No	Yes	No	No
35	UPDATE_EVT	R	DS73	C	Event Update	No	No	No	No
36	BLOCK_ALM	R	DS72	C	Alarm	No	No	No	No
37	ALARM_SUM	R	DS74	C	Alarm Sum	Yes	No	Yes	No

Function Block User Application Reference

Resource Block Parameters

RB Idx	Parameter Name	OT	Data Type	Use	Model	VIEW_x			
						1	2	3	4
38	ACK_OPTION	S	BS	C	Contained	No	No	No	Yes
39	WRITE_PRI	S	US8	C	Alert Pri	No	No	No	Yes
40	WRITE_ALM	R	DS72	C	Alarm	No	No	No	No

2.3 Transducer Block Parameters

The following table lists the FF transducer block parameters.



ATTENTION

Note that some of these parameters are for more complex blocks than are currently supported in this release of fieldbus.

Parameter Name	OT	Data Type	Use	Model	VIEW_x			
					1	2	3	4
ACT_FAIL_ACTION	S	US8	C		No	No	No	Yes
ACT_MAN_ID	S	US16	C		No	No	No	Yes
ACT_MODEL_NUM	S	VS	C		No	No	No	No
ACT_SN	S	US32	C		No	No	No	Yes
ALERT_KEY	S	US8	C	Alert Key	No	No	No	Yes
BLOCK_ALM	R	DS72	C	Alarm	No	No	No	No
BLOCK_ERR	S	BS	C	Block Error	Yes	No	Yes	No
BODY_TEMP	R	DS65	C		No	No	Yes	No
BODY_TEMP_UNIT	S	US16	C		No	No	No	Yes
CAL_MIN_SPAN	S	F	C		No	No	No	Yes
CAL_MIN_SPAN_1	S	F	C		No	No	No	Yes
CAL_MIN_SPAN_2	S	F	C		No	No	No	Yes
CAL_POINT_HI	S	F	C		No	Yes	No	No
CAL_POINT_HI_1	S	F	C		No	Yes	No	No
CAL_POINT_HI_2	S	F	C		No	Yes	No	No
CAL_POINT_LO	S	F	C		No	Yes	No	No
CAL_POINT_LO_1	S	F	C		No	Yes	No	No
CAL_POINT_LO_2	S	F	C		No	Yes	No	No

Function Block User Application Reference

Transducer Block Parameters

Parameter Name	OT	Data Type	Use	Model	VIEW_x			
					1	2	3	4
CAL_PROPOGATION_VEL	S	F	C		No	No	No	Yes
CAL_UNIT	S	US16	C		No	No	No	Yes
CAL_UNIT_1	S	US16	C		No	No	No	Yes
CAL_UNIT_2	S	US16	C		No	No	No	Yes
CHANNEL_MAP	A	US16	C		No	No	No	No
COLLECTION_TYPE	S	US16	C		Yes	Yes	Yes	Yes
ELECT_MAN_ID	S	US16	C		No	No	No	Yes
ELECT_MODEL_NUM	S	VS	C		No	No	No	No
ELECT_SN	S	US32	C		No	No	No	Yes
FINAL_POSITION_VALUE	R	DS65	C		Yes	No	Yes	No
FINAL_VALUE	R	DS65	C		Yes	No	Yes	No
FINAL_VALUE_D	R	DS66	C		Yes	No	Yes	No
FINAL_VALUE_RANGE	R	DS68	C		No	Yes	No	No
LEVEL_OFFSET	S	F	C		No	No	No	Yes
LIN_TYPE	S	US8	C		No	No	No	Yes
MODE_BLK	R	DS69	C	Mode	Yes	No	Yes	No
MODULE_SN	S	US32	C		No	No	No	No
PRIMARY_VALUE	R	DS65	C		Yes	No	Yes	No
PRIMARY_VALUE_1	R	DS65	C		Yes	No	Yes	No
PRIMARY_VALUE_2	R	DS65	C		Yes	No	Yes	No
PRIMARY_VALUE_RANGE	R	DS68	C		No	Yes	No	No
PRIMARY_VALUE_RANGE_1	R	DS68	C		No	Yes	No	No

Function Block User Application Reference
Transducer Block Parameters

Parameter Name	OT	Data Type	Use	Model	VIEW_x			
					1	2	3	4
PRIMARY_VALUE_RANGE_2	R	DS68	C		No	Yes	No	No
PRIMARY_VALUE_TYPE	S	US16	C		No	Yes	No	No
PRIMARY_VALUE_TYPE_1	S	US16	C		No	Yes	No	No
PRIMARY_VALUE_TYPE_2	S	US16	C		No	Yes	No	No
SECONDARY_VALUE	R	DS65	C		Yes	No	Yes	No
SECONDARY_VALUE_UNIT	S	US16	C		No	Yes	No	No
SENSOR_CAL_DATE	S	ToD	C		No	No	No	Yes
SENSOR_CAL_DATE_1	S	ToD	C		No	No	No	Yes
SENSOR_CAL_DATE_2	S	ToD	C		No	No	No	Yes
SENSOR_CAL_LOC	S	VS	C		No	No	No	No
SENSOR_CAL_LOC_1	S	VS	C		No	No	No	No
SENSOR_CAL_LOC_2	S	VS	C		No	No	No	No
SENSOR_CAL_METHOD	S	US8	C		No	No	No	Yes
SENSOR_CAL_METHOD_1	S	US8	C		No	No	No	Yes
SENSOR_CAL_METHOD_2	S	US8	C		No	No	No	Yes
SENSOR_CONNECTION	S	US8	C		No	No	No	Yes
SENSOR_CONNECTION_1	S	US8	C		No	No	No	Yes
SENSOR_CONNECTION_2	S	US8	C		No	No	No	Yes

Function Block User Application Reference
Transducer Block Parameters

Parameter Name	OT	Data Type	Use	Model	VIEW_x			
					1	2	3	4
SENSOR_FILL_FLUID	S	US16	C		No	No	No	Yes
SENSOR_ISOLATOR_MTL	S	US16	C		No	No	No	Yes
SENSOR_OFFSET	S	F	C		No	No	No	Yes
SENSOR_RANGE	R	DS68	C		No	No	No	Yes
SENSOR_RANGE_1	R	DS68	C		No	No	No	Yes
SENSOR_RANGE_2	R	DS68	C		No	No	No	Yes
SENSOR_SN	S	US32	C		No	No	No	Yes
SENSOR_SN_1	S	US32	C		No	No	No	Yes
SENSOR_SN_2	S	US32	C		No	No	No	Yes
SENSOR_TEMP	R	DS65	C		No	No	No	No
SENSOR_TEMP_UNIT	S	US16	C		No	No	No	No
SENSOR_TYPE	S	US16	C		No	No	No	Yes
SENSOR_TYPE_1	S	US16	C		No	No	No	Yes
SENSOR_TYPE_2	S	US16	C		No	No	No	Yes
SERVO_GAIN	S	F	C		No	No	No	Yes
SERVO_RATE	S	F	C		No	No	No	Yes
SERVO_RESET	S	F	C		No	No	No	Yes
ST_REV	S	US16	C	Stat Revision	Yes	Yes	Yes	Yes
STRATEGY	S	US16	C	Strategy	No	No	No	Yes
TAG_DESC	S	OS	C	Tag Desc	No	No	No	No
TRANSDUCER_DIRECTORY	A	US16	C		No	No	No	No
UPDATE_EVT	R	DS73	C	Event Update	No	No	No	No

Parameter Name	OT	Data Type	Use	Model	VIEW_x			
					1	2	3	4
VALVE_MAN_ID	S	US16	C		No	No	No	Yes
VALVE_MODEL_NUM	S	VS	C		No	No	No	No
VALVE_SN	S	US32	C		No	No	No	Yes
VAPOR_TEMP	S	F	C		No	No	No	Yes
VAPOR_TEMP_UNIT	S	US16	C		No	No	No	No
VELOCITY_UNIT	S	US16	C		No	No	No	Yes
XD_ERROR	S	US8	C		Yes	No	Yes	No

2.4 Analog Input Block Parameters

The following table lists the FF AI Function Block Parameters.

RB Idx	Parameter Name	OT	Data Type	Use	VIEW_x			
					1	2	3	4
1	ST_REV	S	US16	C	Yes	Yes	Yes	Yes
2	TAG_DESC	S	OS	C	No	No	No	No
3	STRATEGY	S	US16	C	No	No	No	Yes
4	ALERT_KEY	S	US8	C	No	No	No	Yes
5	MODE_BLK	R	DS69	C	Yes	No	Yes	No
6	BLOCK_ERR	S	BS	C	Yes	No	Yes	No
7	PV	R	DS65	C	Yes	No	Yes	No
8	OUT	R	DS65	O	Yes	No	Yes	No
9	SIMULATE	R	DS82	C	No	No	No	No
10	XD_SCALE	R	DS68	C	No	Yes	No	No
11	OUT_SCALE	R	DS68	C	No	Yes	No	No
12	GRANT_DENY	R	DS70	C	No	Yes	No	No

Function Block User Application Reference

Analog Input Block Parameters

RB Idx	Parameter Name	OT	Data Type	Use	VIEW_x			
					1	2	3	4
13	IO_OPTS	S	BS	C	No	No	No	Yes
14	STATUS_OPTS	S	BS	C	No	No	No	Yes
15	CHANNEL	S	US16	C	No	No	No	Yes
16	L_TYPE	S	US8	C	No	No	No	Yes
17	LOW_CUT	S	...	C	No	No	No	Yes
18	PV_FTIME	S	...	C	No	No	No	Yes
19	FIELD_VAL	R	DS65	C	Yes	No	Yes	No
20	UPDATE_EVT	R	DS73	C	No	No	No	No
21	BLOCK_ALM	R	DS72	C	No	No	No	No
22	ALARM_SUM	R	DS74	C	Yes	No	Yes	No
23	ACK_OPTION	S	BS	C	No	No	No	Yes
24	ALARM_HYS	S	...	C	No	No	No	Yes
25	HI_HI_PRI	S	US8	C	No	No	No	Yes
26	HI_HI_LIM	S	...	C	No	No	No	Yes
27	HI_PRI	S	US8	C	No	No	No	Yes
28	HI_LIM	S	...	C	No	No	No	Yes
29	LO_PRI	S	US8	C	No	No	No	Yes
30	LO_LIM	S	...	C	No	No	No	Yes
31	LO_LO_PRI	S	US8	C	No	No	No	Yes
32	LO_LO_LIM	S	...	C	No	No	No	Yes
33	HI_HI_ALM	R	DS71	C	No	No	No	No
34	HI_ALM	R	DS71	C	No	No	No	No
35	LO_ALM	R	DS71	C	No	No	No	No
36	LO_LO_ALM	R	DS71	C	No	No	No	No

2.5 PID Function Block Parameters

The following table lists the FF PID Function Block Parameters.

RB Idx	Parameter Name	OT	Data Type	Use	VIEW_x			
					1	2	3	4
46	ACK_OPTION	S	BS	C	No	No	No	Yes
47	ALARM_HYS	S	...	C	No	No	No	Yes
45	ALARM_SUM	R	DS74	C	Yes	No	Yes	No
4	ALERT_KEY	S	US8	C	No	No	No	Yes
25	BAL_TIME	S	F	C	No	No	No	Yes
30	BKCAL_HYS	S	F	C	No	No	No	Yes
27	BKCAL_IN	R	DS65	I	No	No	Yes	No
31	BKCAL_OUT	R	DS65	O	No	No	Yes	No
44	BLOCK_ALM	R	DS72	C	No	No	No	No
6	BLOCK_ERR	S	BS	C	Yes	No	Yes	No
17	BYPASS	S	US8	C	No	Yes	No	No
18	CAS_IN	R	DS65	I	Yes	No	Yes	No
13	CONTROL_OPTS	S	BS	C	No	No	No	Yes
64	DV_HI_ALM	R	DS71	C	No	No	No	No
57	DV_HI_LIM	S	F	C	No	No	No	Yes
56	DV_HI_PRI	S	US8	C	No	No	No	Yes
65	DV_LO_ALM	R	DS71	C	No	No	No	No
59	DV_LO_LIM	S	F	C	No	No	No	Yes
58	DV_LO_PRI	S	US8	C	No	No	No	Yes
42	FF_GAIN	S	F	C	No	No	No	Yes
41	FF_SCALE	R	DS68	C	No	No	No	Yes

Function Block User Application Reference

PID Function Block Parameters

RB Idx	Parameter Name	OT	Data Type	Use	VIEW_x			
					1	2	3	4
40	FF_VAL	R	DS65	I	No	No	Yes	No
23	GAIN	S	F	C	No	No	No	Yes
12	GRANT_DENY	R	DS70	C	No	Yes	No	No
61	HI_ALM	R	DS71	C	No	No	No	No
60	HI_HI_ALM	R	DS71	C	No	No	No	No
49	HI_HI_LIM	S	F	C	No	No	No	Yes
48	HI_HI_PRI	S	US8	C	No	No	No	Yes
51	HI_LIM	S	F	C	No	No	No	Yes
50	HI_PRI	S	US8	C	No	No	No	Yes
15	IN	R	DS65	I	No	No	Yes	No
62	LO_ALM	R	DS71	C	No	No	No	No
53	LO_LIM	S	F	C	No	No	No	Yes
63	LO_LO_ALM	R	DS71	C	No	No	No	No
55	LO_LO_LIM	S	F	C	No	No	No	Yes
54	LO_LO_PRI	S	US8	C	No	No	No	Yes
52	LO_PRI	S	US8	C	No	No	No	Yes
5	MODE_BLK	R	DS69	C	Yes	No	Yes	No
9	OUT	R	DS65	O	Yes	No	Yes	No
28	OUT_HI_LIM	S	F	C	No	Yes	No	No
29	OUT_LO_LIM	S	F	C	No	Yes	No	No
11	OUT_SCALE	R	DS68	C	No	Yes	No	No
7	PV	R	DS65	C	Yes	No	Yes	No
16	PV_FTIME	S	F	C	No	No	No	Yes
10	PV_SCALE	R	DS68	C	No	Yes	No	No

Function Block User Application Reference
PID Function Block Parameters

RB Idx	Parameter Name	OT	Data Type	Use	VIEW_x			
					1	2	3	4
26	RATE	S	F	C	No	No	No	Yes
32	RCAS_IN	R	DS65	C	No	No	Yes	No
35	RCAS_OUT	R	DS65	C	No	No	Yes	No
24	RESET	S	F	C	No	No	No	Yes
33	ROUT_IN	R	DS65	C	No	No	Yes	No
36	ROUT_OUT	R	DS65	C	No	No	Yes	No
34	SHED_OPT	S	US8	C	No	No	No	Yes
8	SP	R	DS65	C	Yes	No	Yes	No
21	SP_HI_LIM	S	F	C	No	Yes	No	No
22	SP_LO_LIM	S	F	C	No	Yes	No	No
19	SP_RATE_DN	S	F	C	No	No	No	Yes
20	SP_RATE_UP	S	F	C	No	No	No	Yes
1	ST_REV	S	US16	C	Yes	Yes	Yes	Yes
14	STATUS_OPTS	S	BS	C	No	No	No	Yes
3	STRATEGY	S	US16	C	No	No	No	Yes
2	TAG_DESC	S	OS	C	No	No	No	No
38	TRK_IN_D	R	DS66	I	Yes	No	Yes	No
37	TRK_SCALE	R	DS68	C	No	No	No	Yes
39	TRK_VAL	R	DS65	I	Yes	No	Yes	No
43	UPDATE_EVT	R	DS73	C	No	No	No	No

2.6 Analog Output Function Block Parameters

The following table lists the FF Analog Output Function Block Parameters.

RB Idx	Parameter Name	OT	Data Type	Use	VIEW_x			
					1	2	3	4
1	ST_REV	S	US16	C	Yes	Yes	Yes	Yes
2	TAG_DESC	S	OS	C	No	No	No	No
3	STRATEGY	S	US16	C	No	No	No	Yes
4	ALERT_KEY	S	US8	C	No	No	No	Yes
5	MODE_BLK	R	DS69	C	Yes	No	Yes	No
6	BLOCK_ERR	S	BS	C	Yes	No	Yes	No
7	PV	R	DS65	C	Yes	No	Yes	No
8	SP	R	DS65	C	Yes	No	Yes	No
9	OUT	R	DS65	O	Yes	No	Yes	No
10	SIMULATE	R	DS82	C	No	No	No	No
11	PV_SCALE	R	DS68	C	No	Yes	No	No
12	XD_SCALE	R	DS68	C	No	Yes	No	No
13	GRANT_DENY	R	DS70	C	No	Yes	No	No
14	IO_OPTS	S	BS	C	No	No	No	Yes
15	STATUS_OPTS	S	BS	C	No	No	No	Yes
16	READBACK	R	DS65	C	Yes	No	Yes	No
17	CAS_IN	R	DS65	I	Yes	No	Yes	No
18	SP_RATE_DN	S	F	C	No	No	No	Yes
19	SP_RATE_UP	S	F	C	No	No	No	Yes
20	SP_HI_LIM	S	F	C	No	Yes	No	No
21	SP_LO_LIM	S	F	C	No	Yes	No	No
22	CHANNEL	S	US16	C	No	No	No	Yes
23	FSTATE_TIME	S	F	C	No	No	No	Yes

Function Block User Application Reference
Analog Output Function Block Parameters

RB Idx	Parameter Name	OT	Data Type	Use	VIEW_x			
					1	2	3	4
24	FSTATE_VAL	S	F	C	No	No	No	Yes
25	BKCAL_OUT	R	DS65	O	No	No	Yes	No
26	RCAS_IN	R	DS65	C	No	No	Yes	No
27	SHED_OPT	S	US8	C	No	No	No	Yes
28	RCAS_OUT	R	DS65	C	No	No	Yes	No
29	UPDATE_EVT	R	DS73	C	No	No	No	No
30	BLOCK_ALM	R	DS72	C	No	No	No	No

Function Block User Application Reference
Analog Output Function Block Parameters

3. Alarms Reference

3.1 Alarm Priority

Alarm functionality

The following table describes the functionality for the alarm priority choices when the Alarm Enable State (ALENBST) = ENABLE.

Action	Description
NOACTION	Alarm is not displayed, printed or journaled.
JOURNAL	Alarm is recorded in the Process Alarm Journal. It is not displayed or printed.
LOW	Alarm is recorded in the Process Alarm Journal, printed on the Real Time Journal (RTJ), and displayed on the alarm Summary display.
HIGH	Alarm is recorded in the Process Alarm Journal, printed on the RTJ, and displayed in the Alarm Summary Displays.
EMERGENCY	Alarm is recorded in the Process Alarm Journal, printed on the RTJ, and displayed on the Alarm Summary Displays.
PRINTER	Alarm is printed on the RTJ. It is not displayed or journaled.
JNLPRINT	Alarm is record in the Process Alarm Journal and printed on the RTJ. It is not displayed.

Correlation FF Alarm Priorities with TPS Network Alarm Priorities

The following table shows the correlation of FF alarm priorities with TPS Network alarm priorities.

FF Alert Priority		TPN Alarm Priority
12-15	Critical	Emergency
8-11	Critical	High
5-7	Advisory	Low
4	Advisory	Printer
3	Advisory	JnlPrint

Alarms Reference

Alarm Parameter Mapping

FF Alert Priority		TPN Alarm Priority
2	Low fixed	Journal
1	No Notification	NoAction
0	No Indication	NoAction

3.2 Alarm Parameter Mapping

Analog Input Alarm Parameter Mapping to IOP and TPN Parameters

The following table shows Analog Input Alarm parameter mapping to IOP and TPN parameters.

FF Source	FBIOP Parameter	TPN Parameter
Hi_Hi_Pri	FFHHPR	PVHHPR
Hi_Hi_Lim	FFHHLM	PVHHTP
Hi_Pri	FFHIPR	PVHIPR
Hi_Lim	FFHILM	PVHITP
Lo_Lim	FFLOLM	PVLOTP
Lo_Pri	FFLOPR	PVLOPR
Lo_Lo_Pri	FFLLPR	PVLLPR
Lo_Lo_Lim	FFLLLM	PVLLTP

PID Block Alarm Parameter Mapping to IOP and TPN Parameters

The following table shows mapping of PID Block Alarm parameters.

FF Source	FBIOP Parameter	TPN Parameter
DV_HI_PRI	FFDVHIPR	DEVHIPRI
DV_HI_LIM	FFDVHILM	DEVHITP
DV_LO_PRI	FFDVLOPR	DEVLOPR
DV_LO_LIM	FFDVLOLM	DEVLOTP

FF Source	FBIOP Parameter	TPN Parameter
HI_HI_PRI	FFHHPR	PVHHPR
HI_HI_LIM	FFHHLM	PVHHTP
HI_PRI	FFHIPR	PVHIPR
HI_LIM	FFHILM	PVHITP
LO_PRI	FFLOPR	PVLOPR
LO_LIM	FFLOLM	PVLOTP
LO_LO_PRI	FFLLPR	PVLLPR
LO_LO_LIM	FFLLLM	PVLLTP
BLOCK_ERR		Slot Soft Failure

3.3 Device and Link Alarms

Block Alarms

The following table lists and defines the Function, Resource, and Transducer block alarms.

Mnemonic	Associated Blocks	Description
FFOTHER	Function, Resource, Transducer	FF Block Alarm: Other
FFBLKCFG	Function, Resource, Transducer	FF Block Alarm: Block Configuration Error
FFLNKCFG	Function, Resource, Transducer	FF Block Alarm: Link Configuration Error
FFSIMACT	Function, Resource, Transducer	FF Block Alarm: Simulate Active
FFLO	Function, Resource, Transducer	FF Block Alarm: Local Override
FFFLSAFE	Function, Resource, Transducer	FF Block Alarm: Device Fail Safe Set

Alarms Reference

Device and Link Alarms

Mnemonic	Associated Blocks	Description
FFDEVNMS	Function, Resource, Transducer	FF Block Alarm: Device Needs Maintenance Soon
FFINFL	Function, Resource, Transducer	FF Block Alarm: Input Failure
FFOUTFL	Function, Resource, Transducer	FF Block Alarm: Output Failure
FFMEMFL	Function, Resource, Transducer	FF Block Alarm: Memory Failure
FFLSTDTA	Function, Resource, Transducer	FF Block Alarm: Lost Static Data
FFLNVDTA	Function, Resource, Transducer	FF Block Alarm: Lost Non Volatile Data
FFRBACKFL	Function, Resource, Transducer	FF Block Alarm: Readback Check Failed
FFDEVNMN	Function, Resource, Transducer	FF Block Alarm: Device Needs Maintenance Now
FFPWRUP	Function, Resource, Transducer	FF Block Alarm: Power-up
FFOOS	Function, Resource, Transducer	FF Block Alarm: Out-Of-Service
TBUNSPEC	Transducer	FF Block Alarm: Transducer Unspecified Error
TBGENRAL	Transducer	FF Block Alarm: Transducer General Error
TBCALERR	Transducer	FF Block Alarm: Transducer Calibration Error
TBCFGERR	Transducer	FF Block Alarm: Transducer Configuration Error
TBELECFL	Transducer	FF Block Alarm: Transducer Electronics Failure
TBMECHFL	Transducer	FF Block Alarm: Transducer Mechanical Failure
TBIOFL	Transducer	FF Block Alarm: Transducer I/O Failure
TBDTAERR	Transducer	FF Block Alarm: Transducer Data Integrity Error
TBSWERR	Transducer	FF Block Alarm: Transducer Software Error

Mnemonic	Associated Blocks	Description
TBALGERR	Transducer	FF Block Alarm: Transducer Algorithm Error

3.4 Standard Process Alarms

Device generated analog alarms

The following table lists the device generated analog alarms.

FF Parameter	TPN Parameter	Associated Flag
LO	PVLO	PVLOFL
HI	PVHI	PVHIFL
LO LO	PVLOLO	PVLLFL
HI HI	PVHIHI	PVHHFL
DV LO	DEVLO	DEVLOFL
DV HI	DEVHI	DEVHIFL

Alert mapping to IOP and TPN parameters

The following table shows the mapping (integration) of alerts to TPN events.

Ord	Enum	Description	UCN Evt Pkt Type	Alm/Evt Type
0	UNDEF	Undefined Alarm	No Action	No Action
1	LO	Low Limit Alarm	10 – Process Alarm	PVLO
2	HI	High Limit Alarm	10 – Process Alarm	PVHI
3	LO LO	Critical Low Limit Alarm	10 – Process Alarm	PVLOLO
4	HI HI	Critical High Limit Alarm	10 – Process Alarm	PVHIHI
5	DV LO	Deviation Low Alarm	10 – Process Alarm	DEVLO
6	DV HI	Deviation High Alarm	10 – Process Alarm	DEVHI
8	BLOCK	Block Alarm: 0	14(5) – System Event	FFOTHER ²
8	BLOCK	Block Alarm: 1	14(5) – System Event	FFBLKCFG ²

Alarms Reference

Standard Process Alarms

Ord	Enum	Description	UCN Evt Pkt Type	Alm/Evt Type
8	BLOCK	Block Alarm: 2	14(5) – System Event	FFLNKCFG ²
8	BLOCK	Block Alarm: 3	14(5) – System Event	FFSIMACT ²
8	BLOCK	Block Alarm: 4	14(5) – System Event	FFLO ²
8	BLOCK	Block Alarm: 5	14(5) – System Event	FFFLSAFE ²
8	BLOCK	Block Alarm: 6	14(5) – System Event	FFDEVNMS ²
8	BLOCK	Block Alarm: 7	14(5) – System Event	FFINFL ²
8	BLOCK	Block Alarm: 8	14(5) – System Event	FFOUTFL ²
8	BLOCK	Block Alarm: 9	14(5) – System Event	FFMEMFL ²
8	BLOCK	Block Alarm: 10	14(5) – System Event	FFLSTDTA ²
8	BLOCK	Block Alarm: 11	14(5) – System Event	FFLNVDTA ²
8	BLOCK	Block Alarm: 12	14(5) – System Event	FFRCKFL ²
8	BLOCK	Block Alarm: 13	14(5) – System Event	FFDEVNMN ²
8	BLOCK	Block Alarm: 14	14(5) – System Event	FFPWRUP ²
8	BLOCK	Block Alarm: 15	14(5) – System Event	FFOOS ²
8	BLOCK	Block Alarm: 16	14(5) – System Event	TBUNSPEC ²
8	BLOCK	Block Alarm: 17	14(5) – System Event	TBGENRAL ²
8	BLOCK	Block Alarm: 18	14(5) – System Event	TBCALERR ²
8	BLOCK	Block Alarm: 19	14(5) – System Event	TBCFGERR ²
8	BLOCK	Block Alarm: 20	14(5) – System Event	TBELECFL ²
8	BLOCK	Block Alarm: 21	14(5) – System Event	TBMECHFL ²
8	BLOCK	Block Alarm: 22	14(5) – System Event	TBIOFL ²
8	BLOCK	Block Alarm: 23	14(5) – System Event	TBDTAERR ²
8	BLOCK	Block Alarm: 24	14(5) – System Event	TBSWERR ²
8	BLOCK	Block Alarm: 25	14(5) – System Event	TBALGERR ²
9	UPDATE	TB Static Data Update Event	14(5) – System Event	TBSTCHNG ¹

Ord	Enum	Description	UCN Evt Pkt Type	Alm/Evt Type
9	UPDATE	FB Static Data Update Event	14(5) – System Event	FBSTCHNG ¹
9	UPDATE	RB Static Data Update Event	14(5) – System Event	RBSTCHNG ¹
10	WRITE	Write Protect Change Alarm	14(5) – System Event	RBWPON ¹
10	WRITE	Write Protect Change Alarm	14(5) – System Event	RBWPOFF ¹
11	UPDATE	Link Object Update Event	14(5) – System Event	FFLOCHNG ¹
12	UPDATE	Trend Object Update Event	No Action	No Action

IOP Box Soft Failures

The following table lists the Fieldbus IOP box soft failures.

Mnemonic	Description
PIFAULTY	Bad Personality Image
BSORBOTH	Bad Boot Sector and/or Platform Sector Images in Flash ROM
PSCRC	Bad Cyclic Redundancy Check for Platform Sector Flash ROM Image
ASPLONLY	Bad Application Sector Image in Flash ROM
HIWATER	Stack Hi Water exceeded
SEG1FAIL	Segment 1 problem or LAS failure
SEG2FAIL	Segment 2 problem or LAS failure
SEG3FAIL	Segment 3 problem or LAS failure
SEG4FAIL	Segment 4 problem or LAS failure
S1DGOFLO	Segment 1 Diagnostic Overflow
S2DGOFLO	Segment 2 Diagnostic Overflow
S3DGOFLO	Segment 3 Diagnostic Overflow
S4DGOFLO	Segment 4 Diagnostic Overflow
DEVFAIL	FF Device Failure
PWRUPLOD	IOP has just powered up and stacks are still loading

IOP generated slot soft failures

The following table lists the IOP generated slot (PECM/SECM) soft failures.

Mnemonic	Description
SEGFAIL	Fieldbus segment has failed
VFDMACHG	VFD matching is in progress
VFDMISMA	VFD mismatch
DEVNOTP	Device not present
DEVFAIL	Device has failed (PECM only)

Device generated soft failures

The following table lists the device generated soft failures.

Mnemonic	Description
FFOTHER	FF Block Alarm: Other
FFBLKCFG	FF Block Alarm: Block Configuration Error
FFLNKCFG	FF Block Alarm: Link Configuration Error
FFSIMACT	FF Block Alarm: Simulate Active
FFLO	FF Block Alarm: Local Override
FFFLSAFE	FF Block Alarm: Device Fail Safe Set
FFDEVNMS	FF Block Alarm: Device Needs Maintenance Soon
FFINFL	FF Block Alarm: Input Failure
FFOUTFL	FF Block Alarm: Output Failure
FFMEMFL	FF Block Alarm: Memory Failure
FFLSTDTA	FF Block Alarm: Lost Static Data
FFLNVDTA	FF Block Alarm: Lost Non Volatile Data
FFRCKFL	FF Block Alarm: Readback Check Failed
FFDEVNMN	FF Block Alarm: Device Needs Maintenance Now

Alarms Reference
Standard Process Alarms

Mnemonic	Description
FFPWRUP	FF Block Alarm: Power-up
FFOOS	FF Block Alarm: Out-Of-Service
TBUNSPEC	FF Block Alarm: Transducer Unspecified Error
TBGENRAL	FF Block Alarm: Transducer General Error
TBCALERR	FF Block Alarm: Transducer Calibration Error
TBCFGERR	FF Block Alarm: Transducer Configuration Error
TBELECFL	FF Block Alarm: Transducer Electronics Failure
TBMECHFL	FF Block Alarm: Transducer Mechanical Failure
TBIOFL	FF Block Alarm: Transducer I/O Failure
TBDTAERR	FF Block Alarm: Transducer Data Integrity Error
TBSWERR	FF Block Alarm: Transducer Software Error
TBALGERR	FF Block Alarm: Transducer Algorithm Error

Alarms Reference
Standard Process Alarms

4. Fieldbus IOP

4.1 Software Slot Assignments

The following table lists the fieldbus IOP software slot assignments.

Slot #	Use
0	(Reserved) IOP Box slot
1-31	AI,PID, and AO function blocks (although PECMS may be assigned to these slots, by convention they are used for FBCMs)
119-32	PECMs (resource blocks), function blocks or transducer blocks (FBCMs)
120-123	SECM mapped to links as follows: • Link at position 1 on IOP assigned to Slot 120 • Link at position 2 on IOP assigned to Slot 121 • Link at position 3 on IOP assigned to Slot 122 • Link at position 4 on IOP assigned to Slot 123
124-127	Reserved for future possible IOP redundancy

5. Fieldbus Control Reference

5.1 HPM Failover to Fieldbus Control

Block mode priorities

The Block mode priorities are listed in the following table.

Mode	Priority
O/S (Out of Service)	7-highest
IMan (Initialization Manual)	6
LO (Local Override)	5
Man (Manual)	4
Auto (Automatic)	3
Cas (Cascade)	2
RCas (Remote-Cascade)	1
ROut (Remote-Output)	0 lowest

Conditions for actual modes

The following table lists the conditions by which Actual mode attributes are calculated and the resulting mode.

Input Parameter Status Attributes, Input Values and Resource State	Mode Parameter (Target Attribute Value)					
	O/S	MAN	AUTO	CAS	RCAS	ROUT
Input Status is Good ¹	Yes	Yes	Yes	Yes	Yes	Yes
RCAS IN has BAD status	Yes	Yes	Yes	Yes	Shed ²	Yes
Remote output IN has BAD status	Yes	Yes	Yes	Yes	Yes	Shed ²
Casc input status attribute BAD	O/S	Yes	Yes	Shed ³	Yes	Yes

Input Parameter Status Attributes, Input Values and Resource State	Mode Parameter (Target Attribute Value)					
	<i>O/S</i>	<i>MAN</i>	<i>AUTO</i>	<i>CAS</i>	<i>RCAS</i>	<i>ROUT</i>
Target mode is RCas, Remote-cascade IN status is not good – Initialization Acknowledge; actual mode attribute last execution was not RCas.					Actual mode of last execution or AUTO or shed ⁴	No
Target mode is ROUT; Remote out IN status is not good — (Initialization Acknowledge); actual mode last execution was not Rout						Actual mode last execution or MAN or shed ⁴
Target mode is Cas; cascade input status not good — (Initialization Acknowledge); actual mode last execution was not CAS				Actual mode from last execution or AUTO or MAN	No	No
Primary Input parameter status attribute is BAD or Uncertain ⁵	Yes	Yes	MAN	MAN	MAN	ROUT
a) Option to treat Uncertain as Good b) Will shed based on Shed Option c) Will shed to next permitted mode d) As determined by Shed Option e) Option to treat Uncertain as BAD and bypass not set						

Back-Calculation inputs

The following table lists the back-calculation inputs.

Input Parameter Status Attributes, Input Values and Resource State	Mode Parameter (Target Attribute Value)					
	<i>O/S</i>	<i>MAN</i>	<i>AUTO</i>	<i>CAS</i>	<i>RCAS</i>	<i>ROUT</i>
Back-calculation input status attribute BAD	Yes	IMAN	IMAN	IMAN	IMAN	IMAN
Back-calculation input parameter Good; fault state active; local override, not invited; or init request	Yes	IMAN	IMAN	IMAN	IMAN	IMAN

Tracking enabled

The following table lists tracking enabled.

Input Parameter Status Attributes, Input Values and Resource State	Mode Parameter (Target Attribute Value)					
	<i>O/S</i>	<i>MAN</i>	<i>AUTO</i>	<i>CAS</i>	<i>RCAS</i>	<i>ROUT</i>
Tracking enabled, initiated; track override manual ENABLED	Yes	LO	LO	LO	LO	LO
Tracking enabled, initiated; track override manual DISABLED	Yes	LO	LO	LO	LO	LO
LO = Local Override						

Fault State/Target Mode Change/Resource State

The following table lists the Fault State/Target Mode Change/Resource State.

Input Parameter Status Attributes, Input Values and Resource State	Mode Parameter (Target Attribute Value)					
	<i>O/S</i>	<i>MAN</i>	<i>AUTO</i>	<i>CAS</i>	<i>RCAS</i>	<i>ROUT</i>
Fault State active (in output function block)	Yes	LO	LO	LO	LO	LO

Fieldbus Control Reference
 HPM Failover to Fieldbus Control

Input Parameter Status Attributes, Input Values and Resource State	Mode Parameter (Target Attribute Value)					
	<i>O/S</i>	<i>MAN</i>	<i>AUTO</i>	<i>CAS</i>	<i>RCAS</i>	<i>ROUT</i>
Target mode changed from O/S to another mode since block last executed	N/A	MAN or EM	MAN or EM	MAN or EM	MAN or EM	MAN or EM
Resource stated in resource block parameter resource state is STANDBY	Yes	Yes	Yes	Yes	Yes	Yes
LO = Local Override EM = Evaluated Mode						

6. HPM Fieldbus Point Display Reference

6.1 Group and Detail Displays

Status byte suffix character indications

The block pages use the 16-character string. The following table defines the first and last suffix character of the block page's 16-character string.

Suffix Character	Definition
<i>First Character</i>	
#	Value is bad
?	Value is uncertain
!	Value is good but has unacknowledged alarm condition
*	Value is good but has acknowledged alarm condition
	Otherwise, value is good
<i>Last Character</i>	
^	Value indicates a high limit constraint
v	Value indicates a low limit constraint
@	Value is a constant (constrained in both directions)
	Otherwise, value is not limited

PV status indicators

Indicator	Definition
Red B	Bad PV
Yellow U	PV is uncertain
<blank>	Normal

Output status indicators

Indicator	Definition
Yellow *	Secondary of this point not in cascade mode (TPS points)
Yellow *	Secondary of this point not accepting cascade mode (FF device)
Red *	Secondary of this point has fault state active (FF device)
Green *	Secondary of this point is requesting initialization (FF device)
Red B	Bad Output (FF device only)
Red H	Bad Output and Hi Limit Indication (FF device only)
Red L	Bad Output and Lo Limit Indication (FF device only)
Red C	Bad Output and Constant Indication (FF device only)
Red F	Initiate Fault-State Indication to downstream block (FF device only)
Yellow H	Output at Hi Limit (TPS Points and FF devices)
Yellow ^	Secondary of this point at setpoint high limit or output high limit, or reset action high limited
Yellow L	Output at Lo Limit (TPS points and FF devices)
Yellow v	Secondary of this point at setpoint high limit or output low limit, or reset action low limited
Yellow C	Output Constant Indication (FF device only)
Green A	Initialization Acknowledge Indication to downstream block (FF device only)
<blank>	Normal

Setpoint Status Indicators

Indicator	Definition
Red B	Bad Setpoint
Yellow H	Setpoint High Limit
White H	Setpoint High Limit (FF device)

Indicator	Definition
White L	Setpoint Low Limit
White C	Setpoint Constant (FF device)
P	Target value processing is preset: AM only
R	Target value processing is running: AM only
*	No control action occurring with point in manual mode (setpoint windup HiLo, does not apply to FF devices) Setpoint cannot be increased because of output limiting (windup high) Setpoint cannot be decreased because of output limiting (windup low)
<blank>	Normal

Fieldbus data types

The following fieldbus data types are described here. Supporting tables for the data types and structures are shown in the *PM Family Parameter Reference Dictionary*.

Element Name	Data Type (Index)	Size
<i>DS-65 Data Type Value and Status—Floating Point Structure¹</i>		
Status	Unsigned8 – (5)	1
Value	Float – (8)	4
<i>DS-65 Data Type Value and Status—Floating Point Structure²</i>		
Status	Unsigned8 – (5)	1
Value	Float – (5)	1
<i>DS-67 Data Type Value and Status—Bitstring Structure²</i>		
Status	Unsigned8 – (5)	1
Value	Bitstring – (14)	2

Element Name	Data Type (Index)	Size
f)	Number of Elements 2, referenced size is an 8 bit byte.	
g)	Number of Elements 2	

6.2 Display Mode Handling

Mode management

The following table defines the Group, Detail, and Block page display mode's handling of priority control and dominance.

2 Char	4 Char	Full Name	Priority Order	Control Order	Dominance Order
OS	OOS	Out-of-Service	8	1	8
IM	IMan	Initialization Manual	7	2	-
LO	LO	Local Override	6	3	-
M	Man	Manual	5	4	5
A	Auto	Auto	4	5	1
C	Cas	Cascade	3	6	2
RC	RCas	Remote Cascade	2	7	3
RO	ROut	Remote Output	1	8	4

Configuration Options

Display modes

An FBCM-resident parameter, MODEDOPT, controls the mode display option.

FF MODE (MODEOPT)	Displayed Mode			Description
0	CAS	RCAS	ROUT	Provides the FF mode nomenclature

FF MODE (MODEOPT)	Displayed Mode			Description
1	CAS	BCAS	BCAS	Permits you to configure RCas or ROut mode and call it "BCAS." Note that: - only one of RCas or ROut can be used in a given instance - only one of the two CAS would be utilized
2	CAS	CAS	CAS	Permits you to configure Cas, RCas, or ROut mode and call it "CAS." Note that: - this should be used only in instances where only one of the Cas, RCas, or ROut modes is used - only one of the three CAS would be utilized

Normal operating mode

The **Normal** operating mode is the mode (or combination of modes) to which the target will be set when the operator presses the **NORMAL** key. Normal modes may be any of the following, which will be displayed on the first page of the detail display or on the group display.

2Char	4Char	Meaning
M	Man	Manual
A	Auto	Automatic
C	Cas	Cascade
RC	RCas	Remote Cascade
RO	ROut	Remote Out



ATTENTION

The normal operating mode can never be OOS (out of service) or LO (Local Override).

7. FF Status Byte Display Reference

7.1 FF Status Byte Data Structure

Data structures coding convention

Quality		Sub-status				Limits	
msb	#	#	#	#	#	#	lsb

Quality bits of status byte

Quality Bits		Definition
<i>Decimal</i>	<i>Binary</i>	
0	00	BAD Data Quality
1	01	UNCERTAIN Data Quality
2	10	GOOD (Non-Cascade) Data Quality ¹ <i>For example, used by AI's OUT, DI's OUT_D</i>
3	11	GOOD (Cascade) Data Quality <i>For example, used by PID's OUT, BKCAL_OUT</i>
a) Not used by OPC (Object Linking and Embedding for Process Control) version 1.0		

Sub-status if quality is BAD

Sub-status	Definition
0	Non-Specific
1	Configuration Error
2	Not Connected
3	Device Failure
4	Sensor Failure
5	No Communication, with Last Usable Value
6	No Communication, with no Last Usable Value
7	Out-Of-Service

Sub-status if quality is UNCERTAIN

Sub-status	Definition
0	Non-Specific
1	Last Usable Value
2	Substitute ¹
3	Initial Value ¹
4	Sensor Conversion Not Accurate
5	Engineering Unit Range Violation
6	Sub-Normal
a) Not used by OPC (Object Linking and Embedding for Process Control) version 1.0	

Sub-status if quality is GOOD and non-cascade

Sub-status	Definition
0	Non-Specific ¹
1	Active Block Alarm ¹
2	Active Advisory Alarm ¹
3	Active Critical Alarm ¹
4	Unacknowledged Block Alarm ¹
5	Unacknowledged Advisory Alarm ¹
6	Unacknowledged Critical Alarm ¹
a) Not used by OPC (Object Linking and Embedding for Process Control) version 1.0	

Sub-status if quality is GOOD and cascade

Sub-status	Definition
0	Non-Specific
1	Initialization Acknowledge ¹ (IA)
2	Initialization Request ¹ (IR)
3	Not Invited ¹ (NI)
4	Not Selected ¹ (NS)
5	Do Not Select ¹ (DNS)
6	Local Override (LO)
7	Fault-State Active ¹ (FSA)
8	Initiate Fault-State ¹ (IFS)
a) Not used by OPC (Object Linking and Embedding for Process Control) version 1.0	

7.2 FF Status Byte Characters

FF Status byte suffix character indications

Suffix Character	Definition
First Character	
#	Value is bad
?	Value is uncertain
!	Value is good but has unacknowledged alarm condition ¹
*	Value is good but has acknowledged alarm condition
	Otherwise, value is good
Last Character	
^	Value indicates a high limit constraint
v	Value indicates a low limit constraint
@	Value is a constant (constrained in both directions)

FF Status Byte Display Reference

FF Status Byte Characters

Suffix Character	Definition
	Otherwise, value is not limited
a)	Because operator acknowledgment is console dependent in TPS, the device alarm indication residing in the device itself always appears to be acknowledged in the FF status byte.

Letters following bad symbol

The following table defines the meaning of any letters following a bad (#) symbol.

Letters Following #	Definition
ER	Configuration Error
NC	Not Connected
DF	Device Failure
SF	Sensor Failure
LU	No communication, Last Usable value
CM	No Comm, no last usable value
OS	Out-of-service
<none>	Non-specific

Letter following uncertain symbol

The following table defines the meaning of any letters following an uncertain (?) symbol.

Letters Following #	Definition
L	Last usable
S	Substitute
I	Initial
A	Sensor conversion not Accurate
R	Engineering unit Range violation
N	Sub-Normal

Letters Following #	Definition
<none>	Non-specific

Letter following unacknowledged or acknowledged symbol

The following table defines the meaning of a letter following an unacknowledged (!, unack'd) or acknowledged (*, ack'd) symbol.

Letters Following #	Definition
A	Advisory alarm
B	Block alarm
C	Critical alarm

Letter as first suffix character

The following table defines the meaning of any letters as the first suffix character without a leading symbol.

Letters Following #	Definition
A or AI	Acknowledge Initialization (IA)
R or RI	Request Initialization (IR)
N or NI	Not invited (NI)
S or NS	Not Selected (NS)
L or LO	Local override (LO)
X or FA	Fault-state active (FSA)
F or FI	Initiate Fault-state (IFS)
C or Cnn	Good(Cas) Subcode Reserved by FF

Letter following a slash symbol

The following table defines the meaning of any letters following a slash (/) symbol.

Letters Following #	Definition
/LL	Low Limit constraint
/HL	High Limit constraint
/CN	Constant (constrained in both directions)

7.3 Group and Detail Display Indicators

Group and detail display setpoint status indicators

Indicator	Definition
Red B	Bad Setpoint
Yellow H	Setpoint High Limit
White H	Setpoint High Limit (FF Device)

Indicator	Definition
Yellow L	Setpoint Low Limit
White L	Setpoint Low Limit (FF device)
White C	Setpoint Constant (FF device)
P	Target value processing is preset: AM only
R	Target value processing is running: AM only
*	No control action occurring with point in manual mode (setpoint windup HiLo, Does not apply to FF devices) Setpoint cannot be increased because of output limiting (windup high) Setpoint cannot be decreased because of output limiting (windup low)
<blank>	Normal

Group and detail display PV status indicators

Indicator	Definition
Red B	Bad PV
Yellow U	PV is Uncertain
<blank>	Normal

Group and detail display output status indicators

Indicator	Definition
Yellow *	Secondary of this point not in cascade mode (TPS points)
Yellow *	Secondary of this point not accepting cascade mode (FF device)
Red *	Secondary of this point has fault state active (FF device)
Green *	Secondary of this point is requesting initialization (FF device)
Red B	Bad Output (FF device only)
Red H	Bad Output and Hi Limit Indication (FF device only)
Red L	Bad Output and Lo Limit Indication (FF device only)

FF Status Byte Display Reference
Group and Detail Display Indicators

Indicator	Definition
Red C	Bad Output and Constant Indication (FF device only)
Red F	Initiate Fault-State Indication to downstream block (FF device only)
Yellow H	Output at Hi Limit (TPS points and FF Devices)
Yellow ^	Secondary of this point at setpoint high limit or output high limit, or reset action high limited
Yellow L	Output at Lo Limit (TPS points and FF Devices)
Yellow v	Secondary of this point at setpoint high limit or output low limit, or reset action low limited.
Yellow C	Output Constant Indication (FF device only)
Green A	Initialization Acknowledge Indication to downstream block (FF device only)
<blank>	Normal

FF status byte indicators when Quality is BAD

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
<any>	<any>	#				
Non-specific	No Limit		#	# Bad	red B	BAD
Non-specific	Low Limit		#	# Bad/LL	red L	BAD_L
Non-specific	High Limit		#	# Bad/HL	red H	BAD_H
Non-specific	Constant		#	# Bad/CN	red C	BAD_C
Configuration Error	No Limit		#ER	# Config Error	red B	B_CONFIG
Configuration Error	Low Limit		#ER	# Config Err/LL	red L	B_CNFG_L
Configuration Error	High Limit		#ER	# Config Err/HL	red H	B_CNFG_H
Configuration Error	Constant		#ER	# Config Err/CN	red C	B_CNFG_C
Not Connected	No Limit		#NC	# Not Connected	red B	B_NOTCON
Not Connected	Low Limit		#NC	# Not Connect/LL	red L	B_NCON_L

FF Status Byte Display Reference
Group and Detail Display Indicators

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
Not Connected	High Limit		#NC	# Not Connect/HL	red H	B_NCON_H
Not Connected	Constant		#NC	# Not Connect/CN	red C	B_NCON_C
Device Failure	No Limit		#DF	# Device Failure	red B	B_DEVFL
Device Failure	Low Limit		#DF	# Device Fail/LL	red L	B_DEVF_L
Device Failure	High Limit		#DF	# Device Fail/HL	red H	B_DEVF_H
Device Failure	Constant		#DF	# Device Fail/CN	red C	B_DEVF_C
Sensor Failure	No Limit		#SF	# Sensor Failure	red B	B_SENSOR
Sensor Failure	Low Limit		#SF	# Sensor Fail/LL	red L	B_SENS_L
Sensor Failure	High Limit		#SF	# Sensor Fail/HL	red H	B_SENS_H
Sensor Failure	Constant		#SF	# Sensor Fail/CN	red C	B_SENS_C
No Comm, Last Usable	No Limit		#LU	# NoComm,LUV	red B	B_LUV
No Comm, Last Usable	Low Limit		#LU	# NoComm,LUV/LL	red L	B_LUV_L
No Comm, Last Usable	High Limit		#LU	# NoComm,LUV/HL	red H	B_LUV_H
No Comm, Last Usable	Constant		#LU	# NoComm,LUV/CN	red C	B_LUV_C
No Comm, No Last Usable	No Limit		#CM	# NoComm, NoLast	red B	B_NOCOMM
No Comm, No Last Usable	Low Limit		#CM	# NoComm/LL	red L	B_NOCM_L
No Comm, No Last Usable	High Limit		#CM	# NoComm/HL	red H	B_NOCM_H
No Comm, No Last Usable	Constant		#CM	# NoComm/CN	red C	B_NOCM_C
Out-Of-Service	No Limit		#OS	# Out-Of-Service	red B	B_OOSRVC
Out-Of-Service	Low Limit		#OS	# Out-Of-Serv/LL	red L	B_OOS_L
Out-Of-Service	High Limit		#OS	# Out-Of-Serv/HL	red H	B_OOS_H
Out-Of-Service	Constant		#OS	# Out-Of-Serv/CN	red C	B_OOS_C
a) Detail Display views						

FF Status Byte Display Reference
Group and Detail Display Indicators

FF status byte indicators when Quality is UNCERTAIN

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
<any>	<any>	?				
Non-specific	No Limit		?	? Uncertain	yel U	UNCERTN
Non-specific	Low Limit		? v	? Uncertain/LL	yel L	UNCERT_L
Non-specific	High Limit		? ^	? Uncertain/HL	yel H	UNCERT_H
Non-specific	Constant		? @	? Uncertain/CN	yel C	UNCERT_C
Last Usable	No Limit		?L	? Last Usable	yel U	U_LUV
Last Usable	Low Limit		?Lv	? Last Usable/LL	yel L	U_LUV_L
Last Usable	High Limit		?L^	? Last Usable/HL	yel H	U_LUV_H
Last Usable	Constant		?L@	? Last Usable/CN	yel C	U_LUV_C
Substitute	No Limit		?S	? Substitute	yel S	U_SUBST
Substitute	Low Limit		?Sv	? Substitute/LL	yel S	U_SUBS_L
Substitute	High Limit		?S^	? Substitute/HL	yel S	U_SUBS_H
Substitute	Constant		?S@	? Substitute/CN	yel S	U_SUBS_C
Initial	No Limit		?I	? Initial	yel U	U_INIT
Initial	Low Limit		?Iv	? Initial/LL	yel L	U_INIT_L
Initial	High Limit		?I^	? Initial/HL	yel H	U_INIT_H
Initial	Constant		?I@	? Initial/CN	yel C	U_INIT_C
Sens Conv Not Accurate	No Limit		?A	? NotAccurate	yel U	U_CONVRS
Sens Conv Not Accurate	Low Limit		?Av	? NotAccurate/LL	yel L	U_CONV_L
Sens Conv Not Accurate	High Limit		?A^	? NotAccurate/HL	yel H	U_CONV_H
Sens Conv Not Accurate	Constant		?A@	? NotAccurate/CN	yel C	U_CONV_C
Engr Unit Range Violation	No Limit		?R	? RangeViolat	yel U	U_RANGEV
Engr Unit Range Violation	Low Limit		?Rv	? RangeViolat/LL	yel L	U_RNGE_L
Engr Unit Range Violation	High Limit		?R^	? RangeViolat/HL	yel H	U_RNGE_H
Engr Unit Range Violation	Constant		?R@	? RangeViolat/CN	yel C	U_RNGE_C

FF Status Byte Display Reference
Group and Detail Display Indicators

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
Sub-Normal	No Limit		?N	? Sub-Normal	yel U	U_SUBNRM
Sub-Normal	Low Limit		?Nv	? Sub-Normal/LL	yel L	U_SUBN_L
Sub-Normal	High Limit		?N^	? Sub-Normal/HL	yel H	U_SUBN_H
Sub-Normal	Constant		?N@	? Sub-Normal/CN	yel C	U_SUBN_C
a) Detail Display views = Bas ? = Uncertain ! = Unacknowledged Alarm ^ = Acknowledged Alarm v = Low Limit (Lower case "V") @ = Constant						

FF status byte indicators when Quality is GOOD with No Cascade

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
<any ack'd alarm>	<any>	*				
Non-specific	No Limit					
Non-specific	Low Limit	v				
Non-specific	High Limit	^				
Non-specific	Constant	@				
Non-specific	No Limit					GOOD
Non-specific	Low Limit	v	v	Low Limit	wht L	LOWLIMIT
Non-specific	High Limit	^	^	High Limit	wht H	HILIMIT
Non-specific	Constant	@	@	Constant	wht C	CONSTANT
Active Advisory Alarm	No Limit	*	*A	* Advis Alarm		ADVISALM
Active Advisory Alarm	Low Limit	*	*Av	* Advis Alarm/LL	wht L	ADVALM_L
Active Advisory Alarm	High Limit	*	*A^	* Advis Alarm/HL	wht H	ADVALM_H
Active Advisory Alarm	Constant	*	*A@	* Advis Alarm/CN	wht C	ADVALM_C

FF Status Byte Display Reference
Group and Detail Display Indicators

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
Active Block Alarm	No Limit	*	*B	* Block Alarm		BLOCKALM
Active Block Alarm	Low Limit	*	*Bv	* Block Alarm/LL	wht L	BLKALM_L
Active Block Alarm	High Limit	*	*B^	* Block Alarm/HL	wht H	BLKALM_H
Active Block Alarm	Constant	*	*B@	* Block Alarm/CN	wht C	BLKALM_C
Active Critical Alarm	No Limit	*	*C	* Crit Alarm		CRITALM
Active Critical Alarm	Low Limit	*	*Cv	* Crit Alarm/LL	wht L	CRTALM_L
Active Critical Alarm	High Limit	*	*C^	* Crit Alarm/HL	wht H	CRTALM_H
Active Critical Alarm	Constant	*	*C@	* Crit Alarm/CN	wht C	CRTALM_C
Unack'd Advisory Alarm*	No Limit	!	!A	! Advis Alarm		UADVALM
Unack'd Advisory Alarm*	Low Limit	!	!Av	! Advis Alarm/LL	wht L	UADALM_L
Unack'd Advisory Alarm*	High Limit	!	!A^	! Advis Alarm/HL	wht H	UADALM_H
Unack'd Advisory Alarm*	Constant	!	!A@	! Advis Alarm/CN	wht C	UADALM_C
Unack'd Block Alarm*	No Limit	!	!B	! Block Alarm		UBLKALM
Unack'd Block Alarm*	Low Limit	!	!Bv	! Block Alarm/LL	wht L	UBKALM_L
Unack'd Block Alarm*	High Limit	!	!B^	! Block Alarm/HL	wht H	UBKALM_H
Unack'd Block Alarm*	Constant	!	!B@	! Block Alarm/CN	wht C	UBKALM_C
Unack'd Critical Alarm*	No Limit	!	!C	! Crit Alarm		UCRITALM
Unack'd Critical Alarm*	Low Limit	!	!Cv	! Crit Alarm/LL	wht L	UCRALM_L
Unack'd Critical Alarm*	High Limit	!	!C^	! Crit Alarm/HL	wht H	UCRALM_H
Unack'd Critical Alarm*	Constant	!	!C@	! Crit Alarm/CN	wht C	UCRALM_C
a) Detail Display views = Bas ? = Uncertain ! = Unacknowledged Alarm ^ = Acknowledged Alarm v = Low Limit (Lower case "V") @ = Constant						

FF status byte indicators when Quality is GOOD with Cascade

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
Non-specific	No Limit					GOOD_CAS
Non-specific	Low Limit	v	v	Low Limit	wht L	LOWLIMIT
Non-specific	High Limit	^	^	High Limit	wht H	HILIMIT
Non-specific	Constant	@	@	Constant	wht C	CONSTANT
Acknowledge initializ'n (IA)	<any>	A				
Acknowledge initializ'n (IA)	No Limit	A	AI	Init'n Ack	grn A	IA
Acknowledge initializ'n (IA)	Low Limit	A	AIv	Init'n Ack/LL	wht L	IA_L
Acknowledge initializ'n (IA)	High Limit	A	AI^	Init'n Ack/HL	wht H	IA_H
Acknowledge initializ'n (IA)	Constant	A	AI@	Init'n Ack/CN	wht C	IA_C
Request initialization (IR)	<any>	R				
Request initialization (IR)	No Limit	R	RI	Init'n Req	grn R	IR
Request initialization (IR)	Low Limit	R	RIv	Init'n Req/LL	wht L	IR_L
Request initialization (IR)	High Limit	R	RI^	Init'n Req/HL	wht H	IR_H
Request initialization (IR)	Constant	R	RI@	Init'n Req/CN	wht C	IR_C
Not invited (NI)	<any>	N				
Not Invited (NI)	No Limit	N	NI	Not Invited	yel N	NI
Not Invited (NI)	Low Limit	N	NIv	Not Invited/LL	wht L	NI_L
Not Invited (NI)	High Limit	N	NI^	Not Invited/HL	wht H	NI_H
Not Invited (NI)	Constant	N	NI@	Not Invited/CN	wht C	NI_C
not Selected (NS)	<any>	S				
Not Selected (NS)	No Limit	S	NS	Not Selected	wht S	NS
Not Selected (NS)	Low Limit	S	NSv	Not Selected/LL	wht L	NS_L
Not Selected (NS)	High Limit	S	NS^	Not Selected/HL	wht H	NS_H
Not Selected (NS)	Constant	S	NS@	Not Selected/CN	wht C	NS_C
Reserved_SS5	<any>	D	C05	(Used to be "Do Not Select")		

FF Status Byte Display Reference
Group and Detail Display Indicators

Sub-Status	Limits	Character Suffix			Character Independent Field	
		1	3	16 ¹	1	8
Reserved_SS5	No Limit	D	C05	GoodCas_SS05	red D	C_SS05
Reserved_SS5	Low Limit	D	C05	GoodCas_SS05/LL	wht L	C_SS05_L
Reserved_SS5	High Limit	D	C05	GoodCas_SS05/HL	wht H	C_SS05_H
Reserved_SS5	Constant	D	C05	GoodCas_SS05/CN	wht C	C_SS05_C
Local override (LO)	<any>	L				
Local Override (LO)	No Limit	L	LO	Local Ov'ride	yel O	LO
Local Override (LO)	Low Limit	L	LOv	Local Ov'ride/LL	wht L	LO_L
Local Override (LO)	High Limit	L	LO^	Local Ov'ride/HL	wht H	LO_H
Local Override (LO)	Constant	L	LO@	Local Ov'ride/CN	wht C	LO_C
fault-state active (FSA)	<any>	X				
Fault-State Active (FSA)	No Limit	X	FA	F-State Activ	red X	FSA
Fault-State Active (FSA)	Low Limit	X	FAv	F-State Activ/LL	wht L	FSA_L
Fault-State Active (FSA)	High Limit	X	FA^	F-State Activ/HL	wht H	FSA_H
Fault-State Active (FSA)	Constant	X	FA@	F-State Activ/CN	wht C	FSA_C
initiate Fault-state (IFS)	<any>	F				
Initiate Fault-State (IFS)	No Limit	F	FI	Init F-State	red F	IFS
Initiate Fault-State (IFS)	Low Limit	F	FIv	Init F-State/LL	wht L	IFS_L
Initiate Fault-State (IFS)	High Limit	F	FI^	Init F-State/HL	wht H	IFS_H
Initiate Fault-State (IFS)	Constant	F	FI@	Init F-State/CN	wht C	IFS_C
a) Detail Display views = Bas ? = Uncertain ! = Unacknowledged Alarm ^ = Acknowledged Alarm v = Low Limit (Lower case "V") @ = Constant						

7.4 FF \$FBSTATS Enumeration

\$FBSTATS enumeration when the quality code is zero

The following table defines the \$FBSTATS enumeration when the quality code is one (0) as functions in TPS Release 610.

Sub-status	Base				Limit			
		+1	+2	+3	<i>0 no limit</i>	<i>1 lo limit</i>	<i>2 hi limit</i>	<i>3 constant</i>
0	0	1	2	3	BAD	BAD_L	BAD_H	BAD_C
1	4	5	6	7	B_CONFIG	B_CNFG_L	B_CNFG_H	B_CNFG_C
2	8	9	10	11	B_NOTCON	B_NCON_L	B_NCON_H	B_NCON_C
3	12	13	14	15	B_DEVFL	B_DEVF_L	B_DEVF_H	B_DEVF_C
4	16	17	18	19	B_SENSOR	B_SENS_L	B_SENS_H	B_SENS_C
5	20	21	22	23	B_LUV	B_LUV_L	B_HUV_H	B_CUV_C
6	24	25	26	27	B_NOCOMM	B_NOCM_L	B_NOCM_H	B_NOCM_C
7	28	29	30	31	B_OOSRVC	B_OOS_L	B_OOS_H	B_OOS_C
8	32	33	34	35	B_SS08	B_SS08_L	B_SS08_H	B_SS08_C
9	36	37	38	39	B_SS09	B_SS09_L	B_SS09_H	B_SS09_C
10	40	41	42	43	B_SS10	B_SS10_L	B_SS10_H	B_SS10_C
11	44	45	46	47	B_SS11	B_SS11_L	B_SS11_H	B_SS11_C
12	48	49	50	51	B_SS12	B_SS12_L	B_SS12_H	B_SS12_C
13	52	53	54	55	B_SS13	B_SS13_L	B_SS13_H	B_SS13_C
14	56	57	58	59	B_SS14	B_SS14_L	B_SS14_H	B_SS14_C
15	60	61	62	63	B_SS15	B_SS15_L	B_SS15_H	B_SS15_C

\$FBSTATS enumeration when the quality code is one

The following table defines the \$FBSTATS enumeration when the quality code is one (1) as functions in TPS Release 610.

FF Status Byte Display Reference
FF \$FBSTATS Enumeration

Sub-status	Base				Limit			
		+1	+2	+3	<i>0 no limit</i>	<i>1 lo limit</i>	<i>2 hi limit</i>	<i>3 constant</i>
0	64	65	66	67	UNCERTN	UNCERT_L	UNCERT_H	UNCERT_C
1	68	69	70	71	U_LUV	U_LUV_L	U_HUV_H	U_CUV_C
2	72	73	74	75	U_SUBST	U_SUBS_L	U_SUBS_H	U_SUBS_C
3	76	77	78	79	U_INIT	U_INIT_L	U_INIT_H	U_INIT_C
4	80	81	82	83	U_CONVRS	U_CONV_L	U_CONV_H	U_CONV_C
5	84	85	86	87	U_RANGEV	U_RNGE_L	U_RNGE_H	U_RNGE_C
6	88	89	90	91	U_SUBNRM	U_SUBN_L	U_SUBN_H	U_SUBN_C
7	92	93	94	95	U_SS07	U_SS07_L	U_SS07_H	U_SS07_C
8	96	97	98	99	U_SS08	U_SS08_L	U_SS08_H	U_SS08_C
9	100	101	102	103	U_SS09	U_SS09_L	U_SS09_H	U_SS09_C
10	104	105	106	107	U_SS10	U_SS10_L	U_SS10_H	U_SS10_C
11	108	109	110	111	U_SS11	U_SS11_L	U_SS11_H	U_SS11_C
12	112	113	114	115	U_SS12	U_SS12_L	U_SS12_H	U_SS12_C
13	116	117	118	119	U_SS13	U_SS13_L	U_SS13_H	U_SS13_C
14	120	121	122	123	U_SS14	U_SS14_L	U_SS14_H	U_SS14_C
15	124	125	126	127	U_SS15	U_SS15_L	U_SS15_H	U_SS15_C

\$FBSTATS enumeration when the quality code is two

The following table defines the \$FBSTATS enumeration when the quality code is two (2) as functions in TPS Release 610.

Sub-status	Base				Limit			
		+1	+2	+3	<i>0 no limit</i>	<i>1 lo limit</i>	<i>2 hi limit</i>	<i>3 constant</i>
0	128	129	130	131	GOOD	LOWLIMIT	HILIMIT	CONSTANT
1	132	133	134	135	ADVISALM	ADVALM_L	ADVALM_H	ADVALM_C
2	136	137	138	139	BLOCKALM	BLKALM_L	BLKALM_H	BLKALM_C
3	140	141	142	143	CRITALM	CRTALM_L	CRTALM_H	CRTALM_C
4	144	145	146	147	UADVALM	UADALM_L	UADALM_H	UADALM_C
5	148	149	150	151	UBLKALM	UBKALM_L	UBKALM_H	UBKALM_C
6	152	153	154	155	UCRITALM	UCRALM_L	UCRALM_H	UCRALM_C
7	156	157	158	159	G_SS07	G_SS07_L	G_SS07_H	G_SS07_C
8	160	161	162	163	G_SS08	G_SS08_L	G_SS08_H	G_SS08_C
9	164	165	166	167	G_SS09	G_SS09_L	G_SS09_H	G_SS09_C
10	168	169	170	171	G_SS10	G_SS10_L	G_SS10_H	G_SS10_C
11	172	173	174	175	G_SS11	G_SS11_L	G_SS11_H	G_SS11_C
12	176	177	178	179	G_SS12	G_SS12_L	G_SS12_H	G_SS12_C
13	180	181	182	183	G_SS13	G_SS13_L	G_SS13_H	G_SS13_C
14	184	185	186	187	G_SS14	G_SS14_L	G_SS14_H	G_SS14_C
15	188	189	190	191	G_SS15	G_SS15_L	G_SS15_H	G_SS15_C

FF Status Byte Display Reference
FF \$FBSTATS Enumeration

\$FBSTATS enumeration when the quality code is three

The following table defines the \$FBSTATS enumeration when the quality code is three (3) as functions in TPS Release 610.

Sub-status	Base				Limit			
		+1	+2	+3	<i>0 no limit</i>	<i>1 lo limit</i>	<i>2 hi limit</i>	<i>3 constant</i>
0	192	193	194	195	GOOD_CAS	LOWLIMIT	HILIMIT	CONSTANT
1	196	197	198	199	IA	IA_L	IA_H	IA_C
2	200	201	202	203	IR	IR_L	IR_H	IR_C
3	204	205	206	207	NI	NI_L	NI_H	NI_C
4	208	209	210	211	NS	NS_L	NS_H	NS_C
5	212	213	214	215	C_SS05	C_SS05_L	C_SS05_H	C_SS05_C
6	216	217	218	219	LO	LO_L	LO_H	LO_C
7	220	221	222	223	FSA	FSA_L	FSA_H	FSA_C
8	224	225	226	227	IFS	IFS_L	IFS_H	IFS_C
9	228	229	230	231	C_SS09	C_SS09_L	C_SS09_H	C_SS09_C
10	232	233	234	235	C_SS10	C_SS10_L	C_SS10_H	C_SS10_C
11	236	237	238	239	C_SS11	C_SS11_L	C_SS11_H	C_SS11_C
12	240	241	242	243	C_SS12	C_SS12_L	C_SS12_H	C_SS12_C
13	244	245	246	247	C_SS13	C_SS13_L	C_SS13_H	C_SS13_C
14	248	249	250	251	C_SS14	C_SS14_L	C_SS14_H	C_SS14_C
15	252	253	254	255	C_SS15	C_SS15_L	C_SS15_H	C_SS15_C

8. Parameter Cross-Reference

8.1 TPS to FF Parameter Cross-Reference

TDC Parameter	IOP Data Length	IndexMax	FF Parameter Source
\$BKCLLOUD	2	0	BKCAL_OUT_D
\$BKCLOUT	5	0	BKCAL_OUT
\$BKCLSL1	5	0	BKCAL_SEL1
\$BKCLSL2	5	0	BKCAL_SEL2
\$BKCLSL3	5	0	BKCAL_SEL3
\$RCASOUD	2	0	RCAS_OUT_D
\$RCASOUT	5	0	RCAS_OUT
\$ROUTOUT	5	0	ROUT_OUT
CASIN	5	0	CAS_IN
CASIND	2	0	CAS_IN_D
CVEUHI	4	0	PID: OUT_SCALE.EU100
CVEULO	4	0	PID: OUT_SCALE.EU0
DEVHIFL	1	0	ALARM_SUM.Current
DEVLOFL	1	0	ALARM_SUM.Current
FFALMCUR	0	0	ALARM_SUM.Current
FFBCINST	0	0	BKCAL_IN.Status
FFBCINVL	0	0	BKCAL_IN.Value
FFBCOPST	0	0	BKCAL_OUT.Status
FFBCOPVL	0	0	BKCAL_OUT.Value
FFCSIDST	0	0	CAS_IN_D.Status
FFCSIDVL	0	0	CAS_IN_D.Value

Parameter Cross-ReferenceTPS to FF Parameter Cross-Reference

TDC Parameter	IOP Data Length	IndexMax	FF Parameter Source
FFCSINST	0	0	CAS_IN.Status
FFCSINVL	0	0	CAS_IN.Value
FFDVHILM	0	0	DV_HI_LIM
FFDVHIPR	0	0	DV_HI_PRI
FFDVLOLM	0	0	DV_LO_LIM
FFDVLOPR	0	0	DV_LO_PRI
FFFFDECP	0	0	FF_SCALE.DecPt
FFFFGAIN	0	0	FF_GAIN
FFFFS0	0	0	FF_SCALE.EU0
FFFFS100	0	0	FF_SCALE.EU100
FFFFUIDX	0	0	FF_SCALE.UnitIndex
FFFFVLST	0	0	FF_VAL.Status
FFFFVLVL	0	0	FF_VAL.Value
FFGAIN	0	0	GAIN
FFHHLM	0	0	HI_HI_LIM
FFHHPR	0	0	HI_HI_PRI
FFHILM	0	0	HI_LIM
FFHIPR	0	0	HI_PRI
FFINSTS	0	0	IN.Status
FFINVAL	0	0	IN.Value
FFLLLM	0	0	LO_LO_LIM
FFLLPR	0	0	LO_LO_PRI
FFLOLM	0	0	LO_LIM
FFLOPR	0	0	LO_PRI

Parameter Cross-Reference
TPS to FF Parameter Cross-Reference

TDC Parameter	IOP Data Length	IndexMax	FF Parameter Source
FFMODEA	1	0	MODE_BLK.Actual
FFMODEN	1	0	MODE_BLK.Normal
FFMODEP	1	0	MODE_BLK.Permitted
FFMODET	1	0	MODE_BLK.Target
FFOEXPST	0	0	OUT_EXP.Status
FFOEXPVL	0	0	OUT_EXP.Value
FFOPDECP	0	0	OUT_SCALE.DecPt
FFOPHILM	0	0	OUT_HI_LIM
FFOPLOLM	0	0	OUT_LO_LIM
FFOPS0	0	0	OUT_SCALE.EU0
FFOPS100	0	0	OUT_SCALE.EU100
FFOPUIDX	0	0	OUT_SCALE.UnitIndex
FFOREMST	0	0	OUT_REM.Status
FFOREMVL	0	0	OUT_REM.Value
FFOUT1ST	0	0	OUT_1.Status
FFOUT1VL	0	0	OUT_1.Value
FFOUT2ST	0	0	OUT_2.Status
FFOUT2VL	0	0	OUT_2.Value
FFOUTAST	0	0	OUT_ACCUM.Status
FFOUTAVL	0	0	OUT_ACCUM.Value
FFOUTDST	0	0	OUT_D.Status
FFOUTDVL	0	0	OUT_D.Value
FFOUTSTS	0	0	OUT.Status
FFOUTVAL	0	0	OUT.Value

Parameter Cross-Reference

TPS to FF Parameter Cross-Reference

TDC Parameter	IOP Data Length	IndexMax	FF Parameter Source
FFPTRPST	0	0	OUT_PTRIP.Status
FFPTRPVL	0	0	OUT_PTRIP.Value
FFPVDECP	0	0	PV_SCALE.DecPt
FFPVS0	0	0	PV_SCALE.EU0
FFPVS100	0	0	PV_SCALE.EU100
FFPVSTS	0	0	PV.Status
FFPVUIDX	0	0	PV_SCALE.UnitIndex
FFPVVAL	0	0	PV.Value
FFRATE	0	0	RATE
FFRBDST	0	0	READBACK_D.Status
FFRBDVL	0	0	READBACK_D.Value
FFRBST	0	0	READBACK.Status
FFRBVL	0	0	READBACK.Value
FFRCIDST	0	0	RCAS_IN_D.Status
FFRCIDVL	0	0	RCAS_IN_D.Value
FFRCINST	0	0	RCAS_IN.Status
FFRCINVL	0	0	RCAS_IN.Value
FFRCOPST	0	0	RCAS_OUT.Status
FFRCOPVL	0	0	RCAS_OUT.Value
FFRESET	0	0	RESET
FFROINST	0	0	ROUT_IN.Status
FFROINVL	0	0	ROUT_IN.Value
FFROOPST	0	0	ROUT_OUT.Status
FFROOPVL	0	0	ROUT_OUT.Value

Parameter Cross-Reference
TPS to FF Parameter Cross-Reference

TDC Parameter	IOP Data Length	IndexMax	FF Parameter Source
FFSEL1ST	0	0	SEL_1.Status
FFSEL1VL	0	0	SEL_1.Value
FFSEL2ST	0	0	SEL_2.Status
FFSEL2VL	0	0	SEL_2.Value
FFSEL3ST	0	0	SEL_3.Status
FFSEL3VL	0	0	SEL_3.Value
FFSPHILM	0	0	SP_HI_LIM
FFSPLOLM	0	0	SP_LO_LIM
FFSPRTDN	0	0	SP_RATE_DN
FFSPRTUP	0	0	SP_RATE_UP
FFSPSTS	0	0	SP.Status
FFSPVAL	0	0	SP.Value
FFSTREV	0	0	ST_REV
FFTRDIST	0	0	TRK_IN_D.Status
FFTRDIVL	0	0	TRK_IN_D.Value
FFTRIPST	0	0	OUT_TRIP.Status
FFTRIPVL	0	0	OUT_TRIP.Value
FFTRVLST	0	0	TRK_VAL.Status
FFTRVLVL	0	0	TRK_VAL.Value
FFXDDECP	0	0	XD_SCALE.DecPt
FFXDS0	0	0	XD_SCALE.EU0
FFXDS100	0	0	XD_SCALE.EU100
FFXDUIDX	0	0	XD_SCALE.UnitIndex
FL	0	16	User configured

Parameter Cross-Reference

TPS to FF Parameter Cross-Reference

TDC Parameter	IOP Data Length	IndexMax	FF Parameter Source
IN	4	8	Configurable via link object
IND	1	8	Configurable via link object
INT	0	16	User configured
MODEBLKC	1	0	MODE_BLK.Actual + MODE_BLK.Target
NN	0	16	User configured
OP	4	0	AO, PID: OUT.Value
OUT	4	8	Configurable via link object
OUTD	1	8	Configurable via link object
OUTDST	1	8	Configurable via link object
OUTST	1	8	Configurable via link object
PV	4	0	AI: OUT.Value; PD, PID: PV.Value
PVAUTO	4	0	AI: OUT.Value
PVAUTOST	1	0	AI: OUT.Status
PV CALC	4	0	AI: FIELD_VAL.Value
PVEUHI	4	0	AI: XD_SCALE.EU0; PID: PV_SCALE.EU100
PVEULO	4	0	AI: XD_SCALE.EU0; PID: PV_SCALE.EU0
PVFL	1	0	DI: OUT_D.Value
PVFORMAT	1	0	AI: OUT_SCALE.DecPt; PID: PV_SCALE.DecPt
PVHHFL	1	0	ALARM_SUM.Current
PVHIFL	1	0	ALARM_SUM.Current
PVLLFL	1	0	ALARM_SUM.Current
PVLOFL	1	0	ALARM_SUM.Current

Parameter Cross-Reference
TPS to FF Parameter Cross-Reference

TDC Parameter	IOP Data Length	IndexMax	FF Parameter Source
PVRAW	4	0	None
PVSTS	1	0	AI: OUT.Status; PID: PV.Status
RCASIN	5	0	RCAS_IN
RCASIND	2	0	RCAS_IN_D
ROUTIN	5	0	ROUT_IN
RSSTATE	1	0	Resource block: RS_STATE
SEL1	5	0	SEL_1
SEL2	5	0	SEL_2
SEL3	5	0	SEL_3
SIMDST	0	0	SIMULATE.Simulate Status
SIMDVL	0	0	SIMULATE.Simulate Value
SIMDXST	0	0	SIMULATE.Transducer Status
SIMDXVL	0	0	SIMULATE.Transducer Value
SIMENBL	0	0	SIMULATE.Simulate En/Disable
SIMSTAT	0	0	SIMULATE.Simulate Status
SIMVAL	0	0	SIMULATE.Simulate Value
SIMXSTAT	0	0	SIMULATE.Transducer Status
SIMXVAL	0	0	SIMULATE.Transducer Value
SO	1	0	DO: OUT_D.Value
SP	4	0	AO, PID: SP.Value
SPFORMAT	1	0	AO, PID: PV_SCALE.DecPt
STR	0	16	User configured
VIEWOUT	4	0	None

Parameter Cross-Reference

TPS to FF Parameter Cross-Reference

9. Fieldbus Data Types and Structures Reference

9.1 Fieldbus Data Types and Structures

Index	Type	Name
1	Data	Boolean
2	Data	Integer 8
3	Data	Integer 16
4	Data	Integer 32
5	Data	Unsigned 8
6	Data	Unsigned 16
7	Data	Unsigned 32
8	Data	Floating Point
9	Data	Visible String
10	Data	Octet String
11	Data	Date
12	Data	Time of Day
13	Data	Time Difference
14	Data	Bit String
21	Data	Time Value
64	Structure	Block
65	Structure	Value & Status – Float
66	Structure	Value & Status – Discrete
67	Structure	Value & Status – Bitstring
68	Structure	Scaling
69	Structure	Mode
70	Structure	Access Permissions

Fieldbus Data Types and Structures Reference

Fieldbus Data Types and Structures

Index	Type	Name
71	Structure	Alarm – Floating
72	Structure	Alarm – Discrete
73	Structure	Event – Update
74	Structure	Alarm – Summary
75	Structure	Alert – Analog
76	Structure	Alert – Discrete
77	Structure	Alert – Update
78	Structure	Trend – Float
79	Structure	Tend – Discrete
80	Structure	Trend – Bitstring
81	Structure	FB Link
82	Structure	Simulate – Float
83	Structure	Simulate – Discrete
84	Structure	Simulate – Bitstring
85	Structure	Test
86	Structure	Action – Instantiate/Delete

Reader Comments

Fax Transmittal		Fax No.: (602) 313-4212
Reader Comments		
To:	Information Development	
From:	Name	Date:
	Title:	
	Company:	
	Address:	
	City:	State: Zip:
	Telephone:	Fax:
TPS Fieldbus Reference, FB40-610		
Release 610, 5/00		
Comments:		
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Honeywell Industrial Automation and Control 16404 North Black Canyon Highway Phoenix, AZ 85053		Fieldbus Knowledge Building Tools 